Song Wu

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Education

Wuhan University, Wuhan, China 2020 - 2023
MS, Computer Science, advised by Prof. Jing Chen ,system security

Xidian University, Xi'an, China 2016 - 2020
BS, Computer Science, Elite program, Binary group leader of CTF Team

Publications

When Ad Networks Misbehave: Understanding Risks of Semi-Drive-By in the Splash Ads Ecosystem
First author. Advisors: Qin Wang and Xueqiang Wang. ACM CCS 2026

Understanding Risks of Semi-Drive-By Splash Ads
First author. Advisor: Qin Wang. IEEE S&P 2026 Poster.

Research Experience

MCP Security & Electron Security and Privacy
Intern @Johns Hopkins University, Advisor: Prof. Yinzhi Cao *June 2026 - Present*

- Discovered 18 vulnerabilities in **Model Context Protocol (MCP)** authorization infrastructure, including multiple **20,000+**-star projects and built static ananlysis tools for detection.
- Built **AI-powered static analysis tools for Electron desktop applications** to automatically find **Local Sensitive Logic Exposure**, uncovering **113 zero-day** exploitable vulnerabilities (**3 CVEs** assigned, **3 Acknowledgements**), including critical flaws in **80,000+** star startup projects and AI-powered tools. [*Aiming for IEEE S&P 2027*]

Large Scale Measurement on Splash Ads Fraud In China

- Find a new form of fraud ads named Semi-Drive-By Ads that ads are triggered not by explicit user actions but by incidental or indirect user interactions
- Present **AdHive**, an realphone-like honeypot that induces and detects Semi-Drive-By Ads behavior with learning-assisted DeepLink analysis on **32,758** apps
- Help a top advertiser recover a loss of **$600,000**. [*Accepted at ACM CCS 2026*]

Kernel-level device spoofing framework on Xiaomi Android 13
Researcher@Alibaba *Apr 2024*

- Modified Android kernel to enable spoofing of unique device identifiers, location, charging status, and file information.
- Bypassed Device Identification Tracking of some top vendors and Crawled a top E-commerce data **500,000** times daily

WeChat Mini-apps device spoofing framework
Researcher@Alibaba *July 2024*

- Built a WeChat mini-apps dynamic instrumentation framework with WeChat debug protocol
- Bypassed Device Identification Tracking of some top vendors and craw their merchandise **1,000,000** times daily.

Research on payment vulnerabilities in iOS games
Researcher@Bytedance *Jun 2022*

- Reverse-engineered Apple payment SDK protocols
- Identified a critical Game Recharge Vulnerability and proposed robust defenses to prevent losses of nearly **$1,406,800**

WORK EXPERIENCE

Alibaba July 2023 - Aug 2025
Software Engineer

ByteDance Jun 2022 - Sep 2022
Intern Engineer

SELECTED VULNERABILITIES

CVE-2026-59971 (High Severity)
- SQL execution in a widely used sql mcp with 1.4k+ stars

CVE-2026-7064(High Severity)
- a widely used browser MCP with 7.3K+ stars

CVE-2026-55629 (High Severity)
- an open-source proxy debugging tool with 15.7K+ stars

CVE-2020-14130 (High Severity)
- One-click in Xiaomi,automatic download and execution of malicious applications similar to pwn2own 2019

SKILLS

Reverse Engineer on multi-arch
- Windows, Android, iOS, Wechat Mini Program, IoT

Vulnerability Research
- Web, MCP, Linux, Android, Node.js, 5G, IOT

Programming
- C/C++, JavaScript, Java, Assembly Language

GameHacking
- cocos2d-lua, cocos2d-js, unity3d, protection measures on game

Static Analysis
- soot, closure-compiler, self-developed taint analysis for javascript

HONORS & AWARDS

Outstanding Individual, Huawei HarmonyOS Security Training Camp	2021
First Prize, Vivo Qianjing Cup Cybersecurity Competition	2021
First Prize, National College Student Information Security Contest (CTF)	2019
First Prize, National College Student Information Security Contest (Project)	2019
Third Place, TCTF University Division	2019
Third Prize, West Lake Cybersecurity Forum (Team Competition)	2018
Second Place, Anheng Monthly Cybersecurity Competition (Individual)	2018
Best Defense Team, China Cybersecurity Technology Competition (AWD)	2018
Third Prize, China Cybersecurity Technology Competition (AI)	2018
2018 Third Prize, Wangding Cup (National Online Competition)	2018